Convex optimization - Project 2

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maximum ramp rate with no storage :

no storage means no battery $(c_t = 0)$ so $g_t = n_t + 0 \Rightarrow g_t = n_t$

So

$$\min \sum_{t=1}^{96} (a g_t + b g_t^2)$$

$$\text{s.t.} \quad g_t = n_t + c_t$$

$$q_{t+1} = q_t + 0.25 c_t \qquad t = 1, \dots, 95$$

$$q_1 = q_{96} + 0.25 c_{96}$$

$$0 \leq q_t \leq Q \qquad t = 1, \dots, 96$$

$$-C \leq c_t \leq C \qquad t = 1, \dots, 96$$

$$4 * \max\{|g_2 - g_1|, \dots, |g_1 - g_{96}|\} \leq$$

$$0.5 \times 4 \times \max\{|n_2 - n_1|, \dots, |n_1 - n_{96}|\}$$

I solved it and it's in Code file.